SOFTWARE CHANGE IMPLEMENTATION REPORT

CR033 / CR-006: Last Login Tracking and Display for Student Accounts

Field	Details
Change Request ID	CR033 / CR-006
Change Request Title	Last Login Tracking and Display for Student Accounts
Target System	Student Management System
Student Name	C. Lavasujan
Student ID	24ICT06
Implementation Status	Implemented & Verified
Date	13 October 2026
Document Version	1.0

1. Change Request Summary

1.1 Business Requirement

The system must capture and persist the date and time whenever a student successfully authenticates, then display this information in a dedicated “Last Login” column on the Administrator Dashboard. For student accounts that have not yet logged in, the interface must show the fallback indicator “Never logged in”. The change must preserve existing student-management capabilities and protect the authentication workflow.

1.2 Existing Behaviour

Before this change, the student-management records did not expose a last-login timestamp on the Administrator Dashboard. The requested enhancement therefore adds timestamp storage to the users table, updates that value only after successful password verification, includes it in student retrieval data, and renders a readable value in the dashboard.

- A nullable last_login field is required so existing accounts can remain without a recorded login timestamp.
- Failed login attempts must not update last_login.
- Student records must include the timestamp from the related user record.
- The Admin Dashboard must display a formatted timestamp or “Never logged in”.

1.3 Requested Change

The implementation must:

- Add a nullable last_login DATETIME column to the users table, defaulting to NULL.
- Update the timestamp after successful authentication and before issuing the JWT token.
- Leave the timestamp unchanged when the username or password is invalid.
- Return last_login with student records by joining students to users.
- Add a Last Login column to the Administrator Dashboard and format the value for display.
- Show “Never logged in” for NULL, undefined, or empty timestamp values.
- Preserve existing search, add, edit, delete, authentication, and password-hash behaviour.
- Use parameterized SQL operations to reduce SQL injection risk.

2. Implementation

2.1 Implementation Summary

CR033 / CR-006 implements end-to-end last-login tracking. The users schema now includes a nullable last_login DATETIME field, and the user model exposes updateLastLogin(userId) using a parameterized query. The authentication controller calls this helper only after bcrypt password verification succeeds and before JWT creation. The student model includes u.last_login in the getAllStudents() query through an INNER JOIN, while DashboardPage.jsx formats timestamps and displays “Never logged in” when no timestamp exists. The supplied source report records all seven verification scenarios as passed.

2.2 Affected Files

File	Change	Purpose
database/schema.sql	Modified	Added nullable last_login DATETIME column to the users table with DEFAULT NULL.
backend/src/models/userModel.js	Modified	Added and exported updateLastLogin(userId), using a parameterized UPDATE query.
backend/src/controllers/authController.js	Modified	Updates last_login only after successful bcrypt password verification and before JWT issuance.
backend/src/models/studentModel.js	Modified	Joins students and users and includes u.last_login in getAllStudents().
frontend/src/pages/DashboardPage.jsx	Modified	Adds the Last Login table column and formats values with a “Never logged in” fallback.

The supplied implementation report identifies five modified files. No additional files were specified.

2.3 Technical Changes

Database Schema and Migration

A nullable last_login DATETIME column with DEFAULT NULL was added to the users table. Existing databases require the following migration if the column is not already present:

```
ALTER TABLE users ADD COLUMN last_login DATETIME NULL DEFAULT NULL;
```

User Model and Authentication Flow

- userModel.js implements updateLastLogin(userId) with UPDATE users SET last_login = NOW() WHERE user_id = ? and supplies userId as a bound parameter.
- authController.js calls updateLastLogin(user.user_id) strictly after bcrypt.compare confirms the password is correct and immediately before token creation.
- Invalid username or password paths return HTTP 401 and do not invoke the timestamp update.

Student Retrieval and Dashboard Display

- studentModel.js uses INNER JOIN users u ON s.user_id = u.user_id and projects u.last_login with the student profile fields.
- DashboardPage.jsx adds a “Last Login” heading between Phone and Actions, with a corresponding cell for each student row.
- formatLastLogin() returns “Never logged in” when the value is null, undefined, or empty; otherwise it renders a localized date-and-time string.

2.4 Implementation Logic

1. A student submits a username and password to the existing login endpoint.
2. The authentication controller retrieves the user and verifies the password hash with bcrypt.
3. If verification fails, the request returns HTTP 401 and the existing last_login value remains unchanged.
4. If verification succeeds, updateLastLogin(user.user_id) stores the current database time.
5. The controller proceeds with JWT creation and the normal successful-login response.
6. The Admin Dashboard fetches student records; getAllStudents() supplies last_login from the related users row.
7. DashboardPage.jsx displays a formatted timestamp or “Never logged in” when the value is missing.

2.5 Error and Data Handling

- Unsuccessful authentication: invalid credentials return HTTP 401; the last-login timestamp is not updated.
- Missing timestamp: NULL or empty values are rendered as “Never logged in”, avoiding blank or invalid-looking output.
- Query safety: last-login updates use a parameterized query with a bound user ID.
- Backward compatibility: existing student search and standard CRUD operations remain available; the report records these regression checks as passed.

2.6 Deployment and Operational Notes

8. Ensure the database migration has been applied on the target environment if last_login is not yet present in users.
9. Restart or reload the Node.js / Express backend so the updated authentication controller and model files are loaded.
10. For production deployment, run npm run build from the frontend directory. In Vite development mode, hot-module replacement may update frontend changes automatically.

3. Testing & Verification

3.1 Testing Approach

The supplied implementation report records the following database, API/authentication, dashboard, and regression scenarios as passed. Exact operating-system, browser-version, and tool-version details were not specified in the source and are therefore not inferred here.

Testing types:

- Functional and positive testing
- Negative authentication testing
- Database timestamp verification
- Dashboard UI verification
- Search regression testing
- Student CRUD regression testing

Environment / Browser: Not specified in the supplied source report.

3.2 Test Cases

Test ID	Test Case	Expected Result	Actual Result	Status
TC-01	New student who has never logged in	Dashboard shows “Never logged in”.	Fallback displayed for NULL value.	Pass
TC-02	Successful student login	last_login is updated; login succeeds and returns JWT.	Timestamp updated; login succeeded.	Pass
TC-03	Incorrect password	HTTP 401; timestamp remains unchanged.	Request rejected; timestamp not updated.	Pass
TC-04	Non-existent username	HTTP 401; no database modification.	Request rejected with 401.	Pass
TC-05	Dashboard display after successful login	Student row shows formatted date and time.	Formatted timestamp rendered in Last Login column.	Pass
TC-06	Client-side search regression	Search filters student rows and preserves Last Login values.	Search worked with last-login values intact.	Pass
TC-07	Student CRUD regression	Add, edit, and delete continue to operate normally.	Existing CRUD operations performed as expected.	Pass

3.3 Test Summary

Test Result	Count
Total Test Cases	7
Passed	7
Failed	0
Blocked	0
Pass Rate	100%

Overall Test Result: Passed (as recorded in the supplied implementation report).

4. Acceptance Criteria Verification

ID	Acceptance Criterion	Verification	Status
FR-001	Nullable last_login DATETIME column is present in users.	Schema review / migration	Met
FR-002	Timestamp updates after successful authentication.	TC-02	Met
FR-003	Failed username/password attempts do not update last_login.	TC-03, TC-04	Met
FR-004	Student retrieval includes last_login from the related users record.	Query / response inspection	Met
FR-005	Administrator Dashboard displays the Last Login column.	TC-05	Met
FR-006	Formatted timestamp or “Never logged in” is shown.	TC-01, TC-05	Met
FR-007	Search, add, edit, and delete remain functional.	TC-06, TC-07	Met
SEC-001	Existing authentication and password-hash protection are preserved.	TC-02 to TC-04 / code review	Met
SEC-002	Parameterized SQL queries are used for last-login update and retrieval.	Query inspection	Met

Verification status above reflects the supplied CR033 / CR-006 implementation report. Any screenshot evidence not yet captured should be inserted in Section 5 before submission.

5. Evidence

Insert the captured screenshots in the evidence areas below. Each screenshot should show the relevant application or test tool clearly and avoid exposing real passwords or unnecessary personal information.

Evidence 01 – Never-Logged-In Fallback

Related: FR-001, FR-004, FR-005, FR-006 / TC-01

Administrator Dashboard showing the Last Login column and “Never logged in” for a student account that has not logged in.

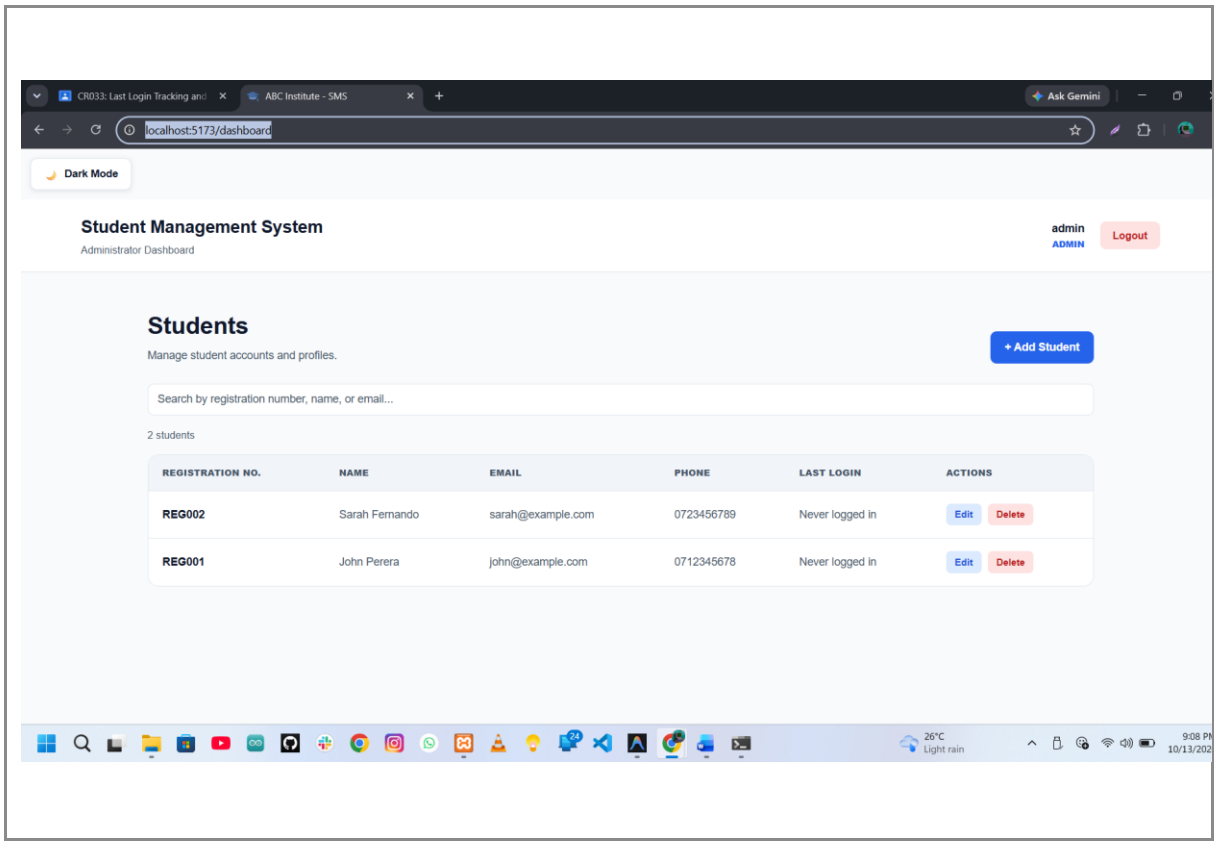


Figure 1: Administrator Dashboard showing the Last Login column and “Never logged in” for a student account that has not logged in.

Evidence 02 – Successful Login Timestamp

Related: FR-002, FR-004, FR-005, FR-006 / TC-02, TC-05

After a successful student login, the Administrator Dashboard shows the same student with a formatted Last Login date and time.

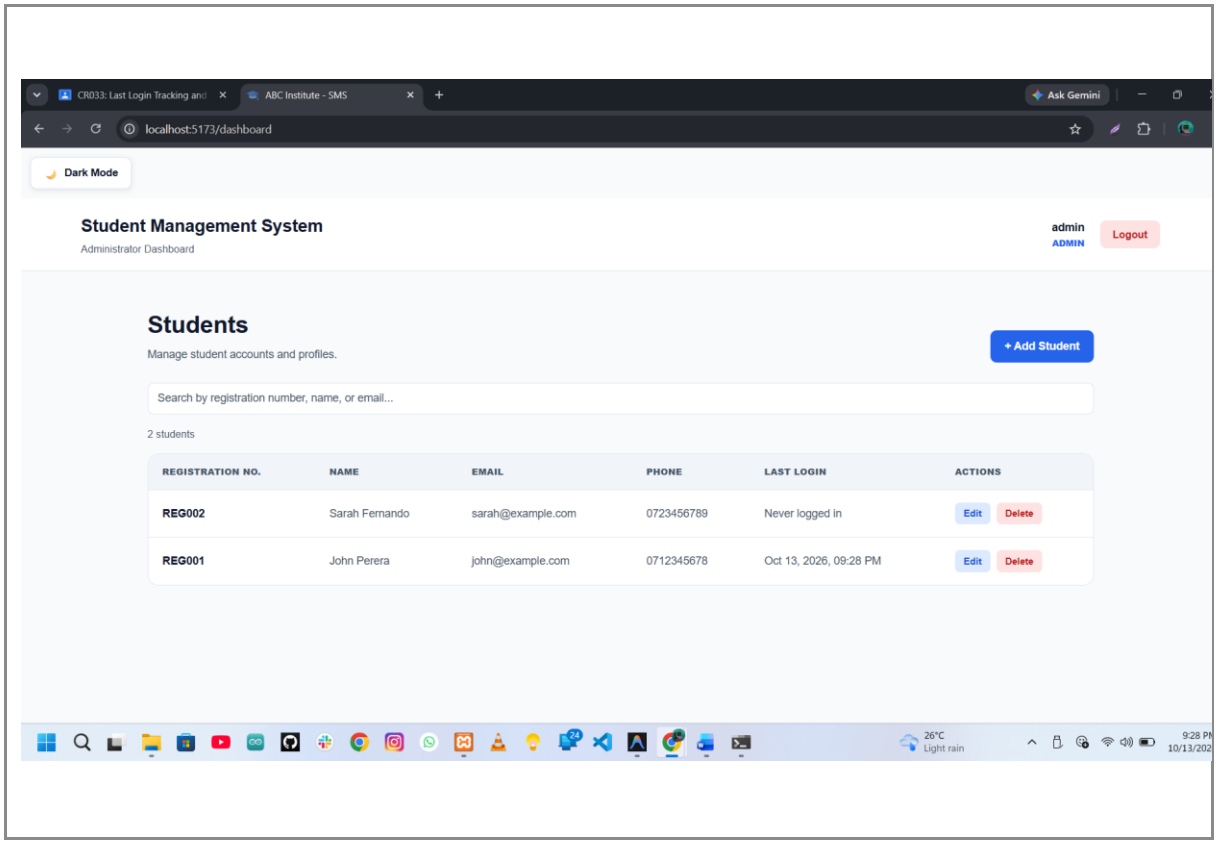


Figure 2: After a successful student login, the Administrator Dashboard shows the same student with a formatted Last Login date and time.

Evidence 03 – Invalid Password Does Not Update Timestamp

Related: FR-003, SEC-001 / TC-03

API client or response log showing HTTP 401 for an incorrect password, together with a database view or query result demonstrating that last_login stayed unchanged.

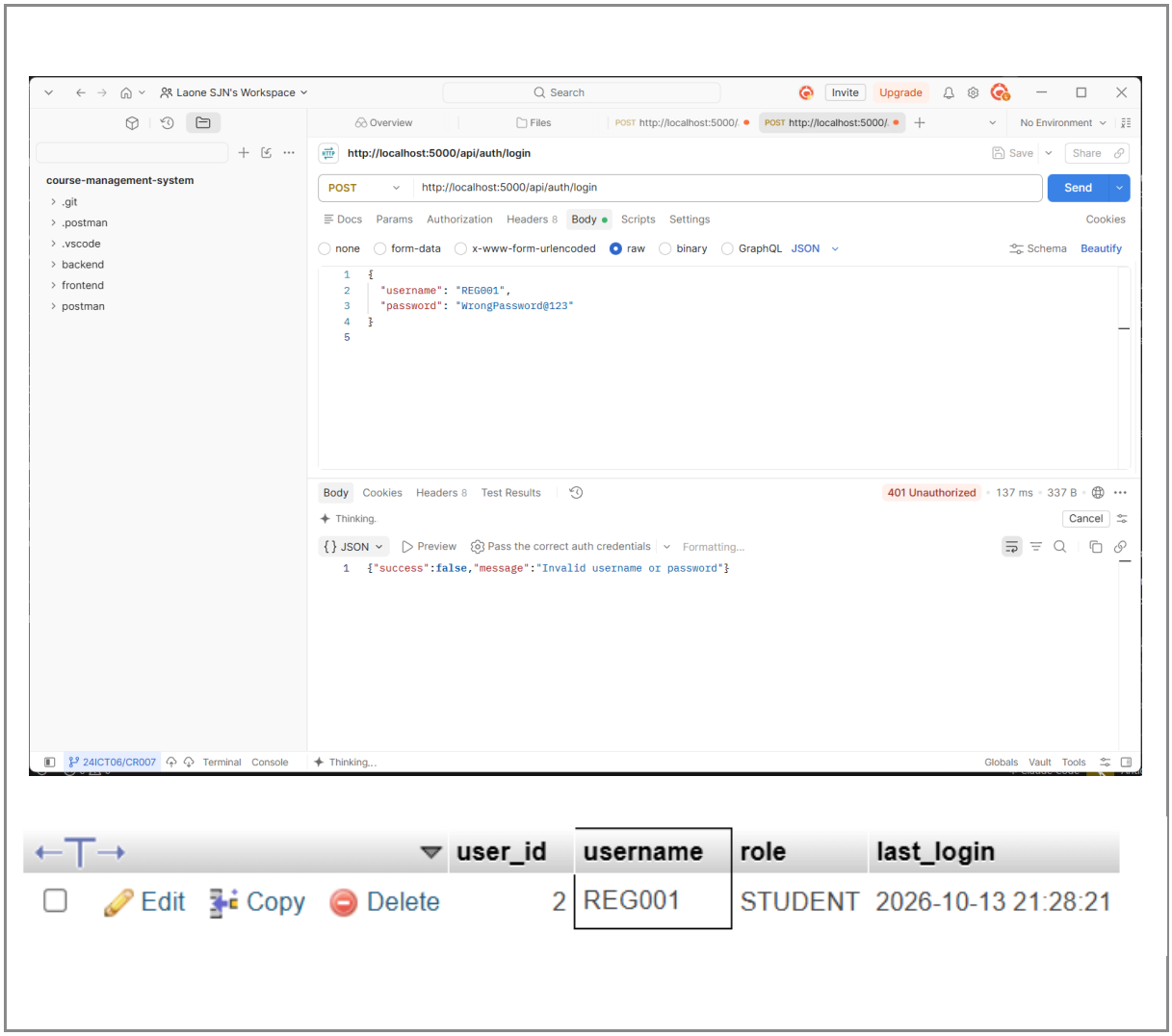


Figure 3: API client or response log showing HTTP 401 for an incorrect password, together with a database view or query result demonstrating that last_login stayed unchanged.

Evidence 04 – Invalid Username Rejection

Related: FR-003, SEC-001 / TC-04

API client or response log showing HTTP 401 for a non-existent username. Do not expose any real account password.

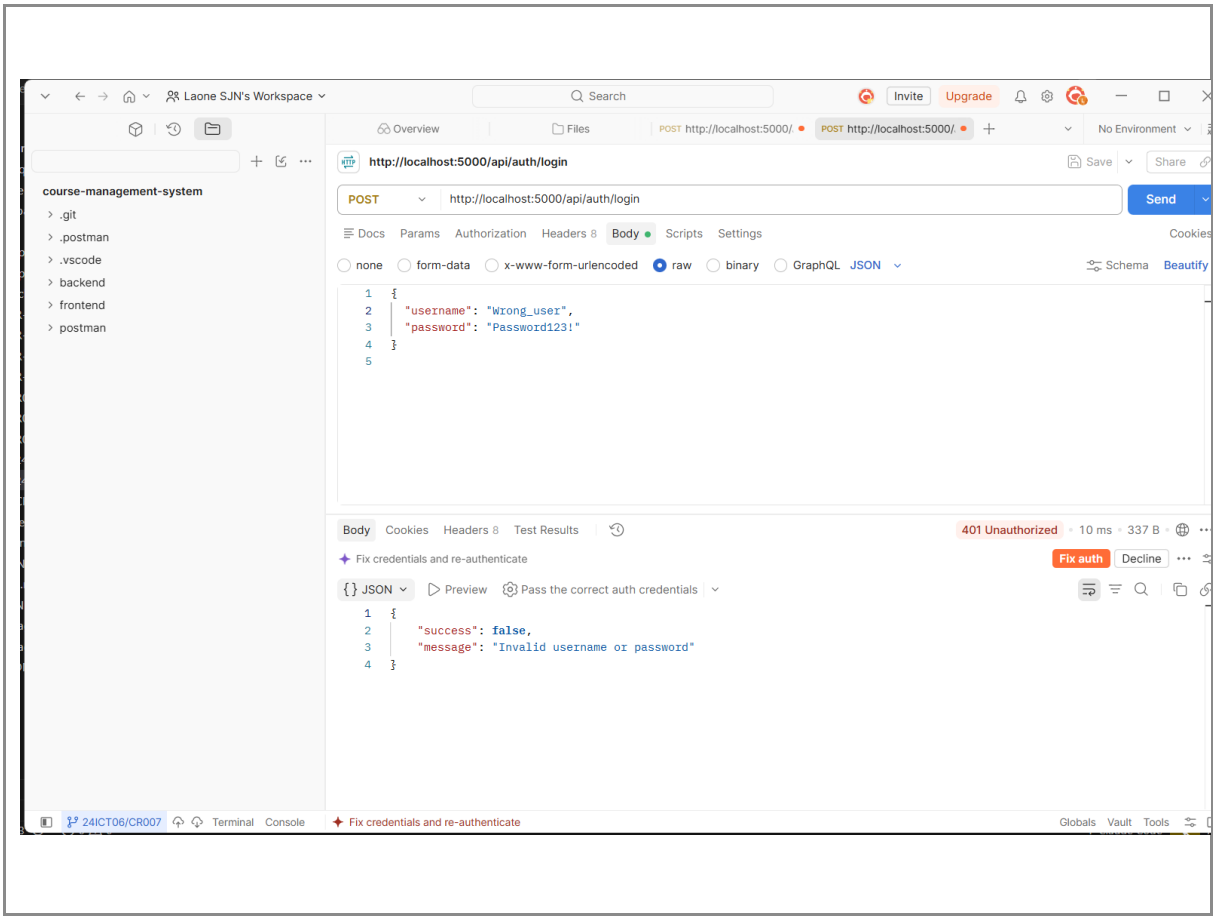


Figure 4: API client or response log showing HTTP 401 for a non-existent username. Do not expose any real account password.

Evidence 05 – Search and CRUD Regression

Related: FR-007 / TC-06, TC-07

Administrator Dashboard showing the Last Login column while search works. Keep supplementary notes or a short test log documenting successful add, edit, and delete verification.

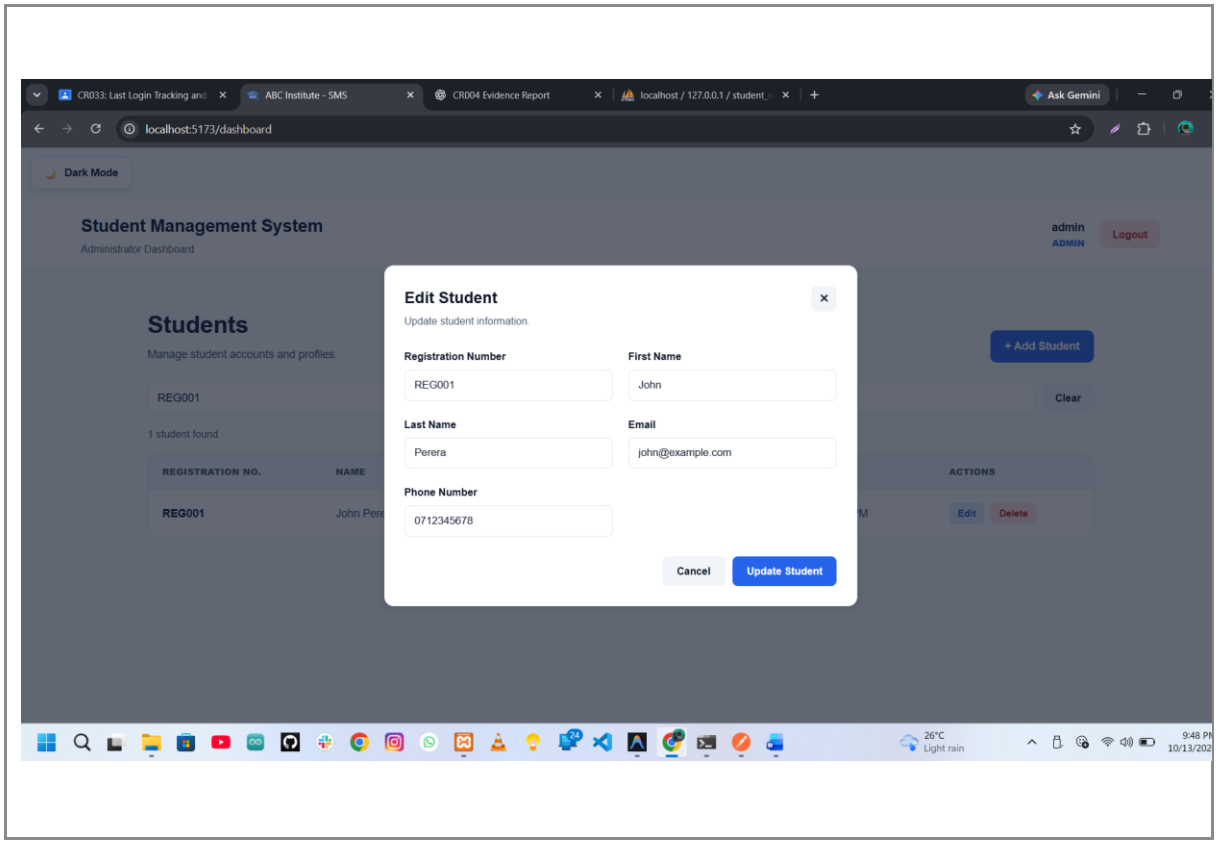


Figure 5: Administrator Dashboard showing the Last Login column while search works. Keep supplementary notes or a short test log documenting successful add, edit, and delete verification.

6. Final Result

Item	Result
Requirements	Implemented / verified in supplied report
Implementation	Completed
Testing	Passed – 7/7 test scenarios
Acceptance Criteria	All listed FR and SEC criteria marked Met
Existing CRUD / Authentication	Preserved and regression verified
Final Status	Implemented & Verified; insert screenshot evidence before submission

6.1 Final Summary

CR033 / CR-006 adds last-login tracking to the Student Management System. The users table stores a nullable last_login timestamp; the authentication flow updates it only after a valid password is verified; the student model returns the value through a join; and the Administrator Dashboard displays either a formatted timestamp or “Never logged in”. The source report states that all seven test scenarios passed and that existing search and CRUD capabilities continue to work. The change requires a database migration on existing installations where the column is not yet present.

6.2 Issues / Limitations

No functional issue is identified in the supplied source report. Before final submission, insert the evidence screenshots listed in Section 5. Before deployment to an existing database, confirm that the last_login migration has been applied. Browser and operating-system versions were not specified in the source report.

Student Name: C. Lavasujan

Student ID: 24ICT06

Date: 13 October 2026

Signature: _____

~ END OF REPORT ~